

CLIMATE CHANGE SCIENCE GREENHOUSE EFFECT

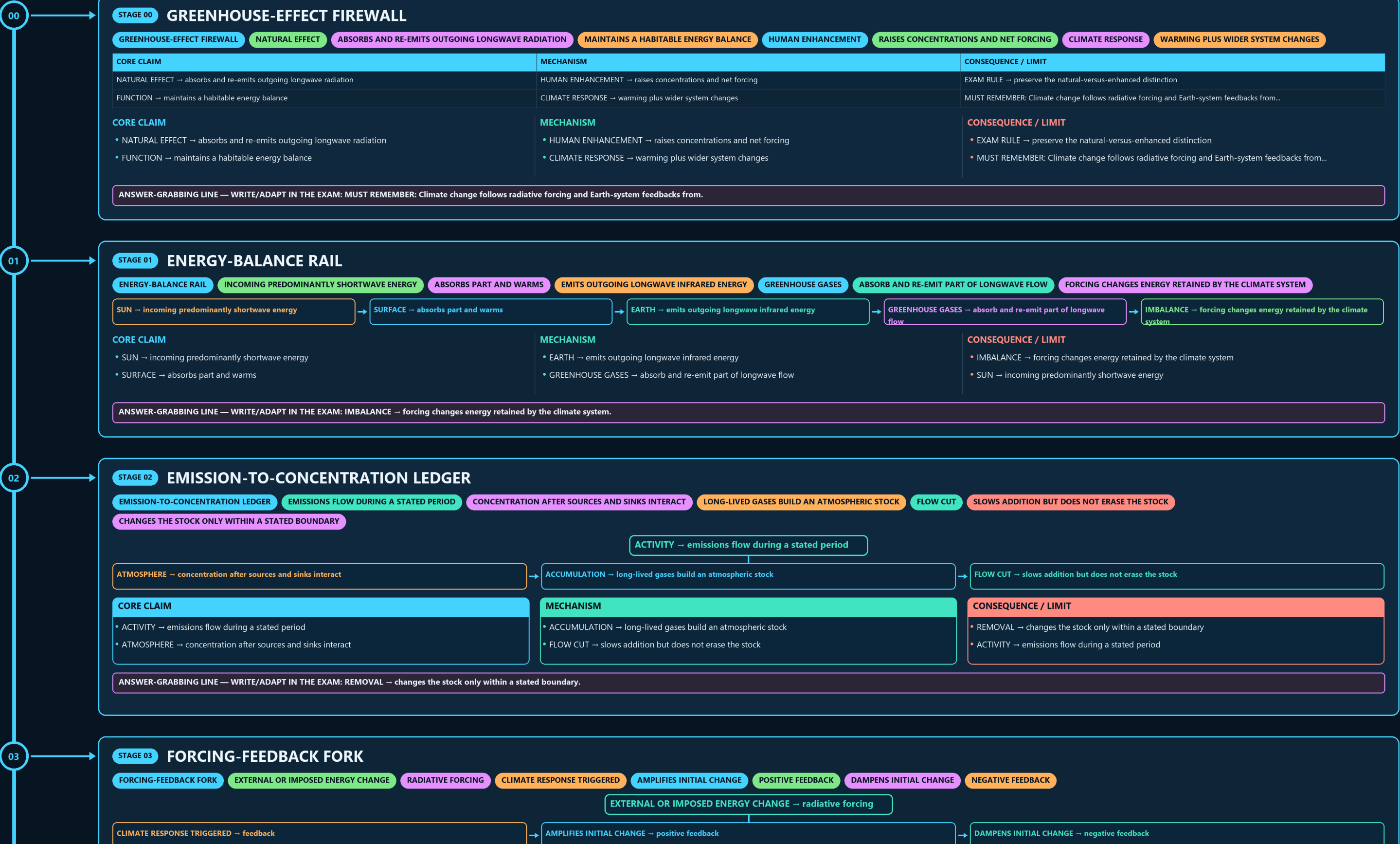
GREENHOUSE-EFFECT FIREWALL → ENERGY-BALANCE RAIL → EMISSION-TO-CONCENTRATION LEDGER → FORCING-FEEDBACK FORK → WATER AND PARTICLE MATRIX → GAS COMPARISON GATE

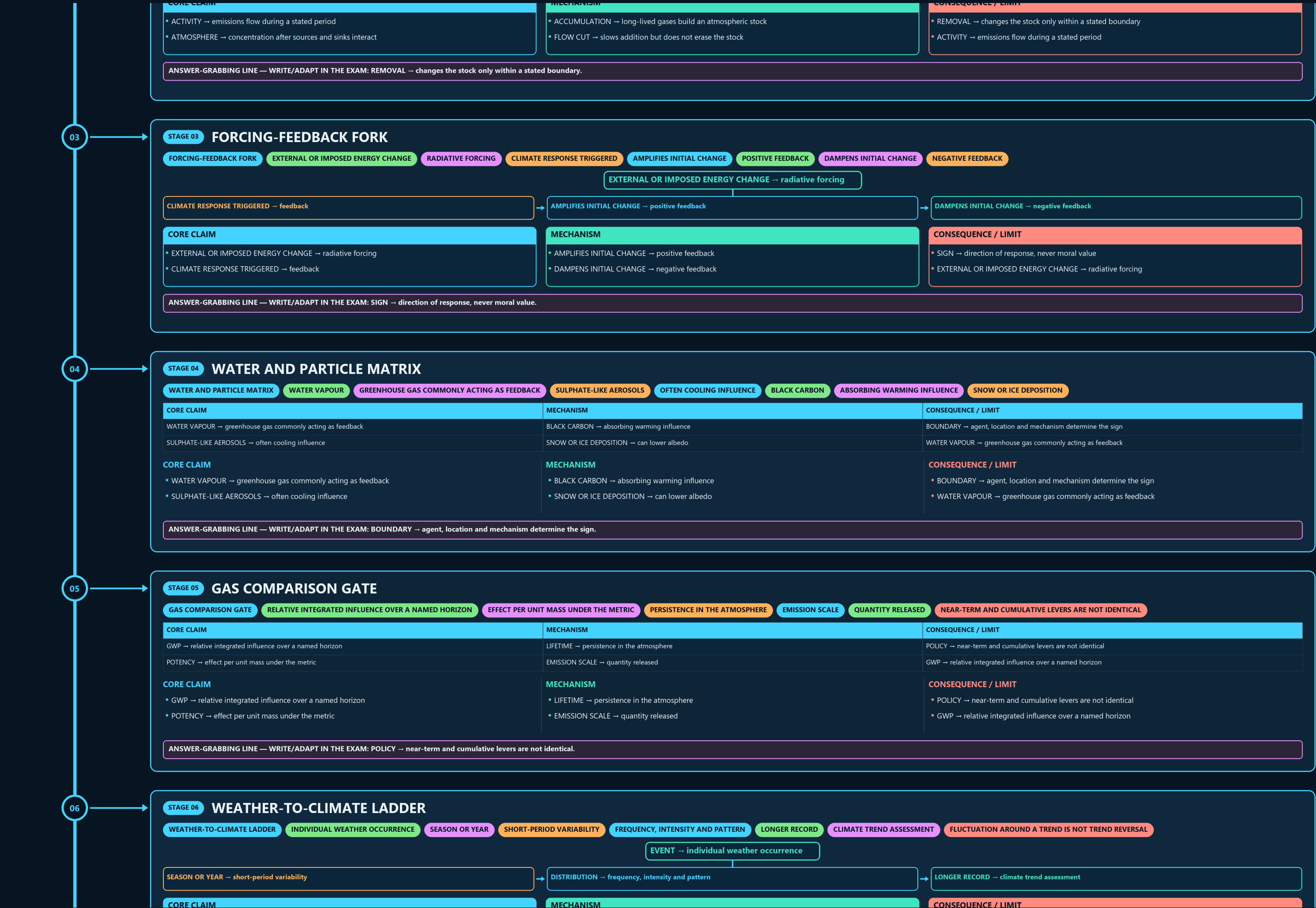
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





- POLICY → near-term and cumulative levers are not identical
- GWP → relative integrated influence over a named horizon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POLICY → near-term and cumulative levers are not identical.

EVENT → individual weather occurrence

- **LONGER RECORD** → climate trend assessment

CONSEQUENCE / LIMIT

- RULE → fluctuation around a trend is not trend reversal
- EVENT → individual weather occurrence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → fluctuation around a trend is not trend reversal.

- **ATTRIBUTION** → qualified contribution, not slogan causation

CONSEQUENCE / LIMIT

- **ATTRIBUTION** → qualified contribution, not slogan causation
- **OBSERVATION** → define the measured change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MODELS PLUS PHYSICS PLUS RECORDS → converging evidence.

CONSEQUENCE / LIMIT

- NO SOLE CAUSE → disaster impact is multi-causal
- DEFINE EVENT CLASS → place, duration and metric

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO SOLE CAUSE → disaster impact is multi-causal.

WEATHER IS NOT CLIMATE, EMISSION FLOW IS NOT CONCENTRATION

CONSEQUENCE / LIMIT

FORECAST → different claim with a prediction horizon

CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...

CONSEQUENCE / LIMIT

- FORECAST → different claim with a prediction horizon
- CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock..

09

STAGE 09

OBSERVATION-PROJECTION MATRIX

OBSERVATION-PROJECTION MATRIX

MEASURED PAST OR PRESENT

EXPLANATION OF OBSERVED CHANGE

CONDITIONAL ASSUMPTIONS

MODELLED RESPONSE UNDER A SCENARIO

DIFFERENT CLAIM WITH A PREDICTION HORIZON

CLOSE DISTINCTION

WEATHER IS NOT CLIMATE, EMISSION FLOW IS NOT CONCENTRATION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
OBSERVATION → measured past or present	SCENARIO → conditional assumptions	FORECAST → different claim with a prediction horizon
ATTRIBUTION → explanation of observed change	PROJECTION → modelled response under a scenario	CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...

CORE CLAIM

OBSERVATION → measured past or present

ATTRIBUTION → explanation of observed change

MECHANISM

SCENARIO → conditional assumptions

PROJECTION → modelled response under a scenario

CONSEQUENCE / LIMIT

FORECAST → different claim with a prediction horizon

CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,.

10

STAGE 10

CLIMATE-RESPONSE FORK

CLIMATE-RESPONSE FORK

SOURCE REDUCTION

SINK ENHANCEMENT

MITIGATION WITH ACCOUNTING BOUNDARY

EXPOSURE OR VULNERABILITY REDUCTION

RESIDUAL HARM

SEPARATE LOSS-AND-DAMAGE QUESTION

MITIGATION AND ADAPTATION ARE COMPLEMENTARY

SOURCE REDUCTION → mitigation

SINK ENHANCEMENT → mitigation with accounting boundary

EXPOSURE OR VULNERABILITY REDUCTION → adaptation

RESIDUAL HARM → separate loss-and-damage question

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SOURCE REDUCTION → mitigation SINK ENHANCEMENT → mitigation with accounting boundary	EXPOSURE OR VULNERABILITY REDUCTION → adaptation RESIDUAL HARM → separate loss-and-damage question	PORTFOLIO → mitigation and adaptation are complementary EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State unit, baseline, period, scenario...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State unit, baseline, period, scenario...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State unit, baseline, period, scenario.

11

STAGE 11

SCIENCE ANSWER SPINE

SCIENCE ANSWER SPINE

NATURAL EFFECT, HUMAN ENHANCEMENT AND ENERGY FLOW

EMISSIONS, CONCENTRATIONS, STOCK AND NET BALANCE

FORCING, FEEDBACK AND GAS-SPECIFIC PROPERTIES

TREND, DETECTION, ATTRIBUTION AND PROJECTION

SCALE, SCENARIO, CONFIDENCE AND NO INVENTED FIGURES

DEFINE → natural effect, human enhancement and energy flow

ACCOUNT → emissions, concentrations, stock and net balance

EXPLAIN → forcing, feedback and gas-specific properties

EVALUATE → trend, detection, attribution and projection

QUALIFY → scale, scenario, confidence and no invented figures

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DEFINE → natural effect, human enhancement and energy flow ACCOUNT → emissions, concentrations, stock and net balance	EXPLAIN → forcing, feedback and gas-specific properties EVALUATE → trend, detection, attribution and projection	QUALIFY → scale, scenario, confidence and no invented figures DEFINE → natural effect, human enhancement and energy flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → scale, scenario, confidence and no invented figures.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S NATIONAL EMISSIONS PROFILE (AS REPORTED IN ITS PERIODIC

ANALYTICAL POINT

INDIA'S PER-CAPITA AND HISTORICAL CUMULATIVE EMISSIONS REMAIN

GLOBAL EMISSIONS AND REMAINING-CARBON-BUDGET FIGURES ARE PERIODICALLY UPDATED BY

ATTRIBUTION-SCIENCE CONFIDENCE LEVELS VARY MEANINGFULLY BY EVENT TYPE (HIGH

INDIA-SPECIFIC SECTORAL EMISSIONS DATA (FROM NATIONAL COMMUNICATIONS/BURS) IS REPORTED

IPCC AR6 ATTRIBUTION SCIENCE HAS SUBSTANTIALLY STRENGTHENED CONFIDENCE IN

OPTIONAL SOURCE DEPTH	ADVANCED DEBATE	LEGACY / NUANCE
CAUTION: India's national emissions profile (as reported in its periodic UNFCCC national CAUTION: Analytical point: India's per-capita and historical cumulative emissions remain CAUTION: Global emissions and remaining-carbon-budget figures are periodically updated by IPCC CAUTION: Attribution-science confidence levels vary meaningfully by event type (high confidence	CAUTION: India-specific sectoral emissions data (from national communications/BURs) is reported IPCC AR6 attribution science has substantially strengthened confidence in linking Aerosols exert a net cooling (negative radiative forcing) effect that partially masks The remaining carbon budget concept reframes climate policy around a finite, depleting	India's methane and nitrous oxide emissions are disproportionately driven by NOT: All extreme weather events can be attributed to climate change with equally high NOT: Reducing air pollution (aerosols) has no bearing on the rate of global warming. - It NOT: A distant net-zero target date alone is sufficient to remain within a fixed carbon

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.